Objectives:

Design a gearbox that can be assembled and disassembled.
Analyze and size input and output shafts for strength.
Design journal bearings to support shafts.
Create a gearbox housing with proper clearance and fastening.
Use parametric CAD modeling to generate a final design.
Document calculations, sketches, and lessons learned.

Assumptions:

Assume appropriate strength of shafts, use Machinery's Handbook.
The gear train is ideal.
The moment of the radial force from the gear train is small enough to be ignored.

Shaft Equations

(length is inches, pressure and force are psi, angular speed is in RPM, Power is in HP)

Main Power Transmitting Shafts

$$P_{xshaft} = \frac{D_{xshaft}^3 N}{80} \text{ or } D_{xshaft} = \left(\frac{80 P_{xshaft}}{N} \right)^{\frac{1}{3}}$$

Lineshafts Carrying Pulleys

$$P_{pulley} = \frac{D_{pulley}^3 N}{53.5} \text{ or } D_{pulley} = \left(\frac{53.5 P}{N} \right)^{\frac{1}{3}}$$

Small, Short Shafts

$$P_{short} = \frac{D_{short}^3 N}{38} \text{ or } D_{short} = \left(\frac{38 P}{N} \right)^{\frac{1}{3}}$$

Gear Ratio

$$G.R. = \frac{Input\ Speed}{Output\ Speed} = \frac{N_I}{N_O}$$

Ideal Gearbox

$$P_I = P_O,\ N_I T_I = N_O T_O,\ \frac{T_O}{T_I} = \frac{N_I}{N_O} = G.R.$$

Polar Section Modulus

$$Z_p = \frac{\pi}{16} D^3$$

Torsional Shearing Stress $\left(S_S\right)$

$$S_S = \frac{T_{ORQUE}}{Z_p}$$

Gear Force Analysis

Tangential Force

$$F_T = \frac{2 \cdot T_{orque}}{d_{pitch}} \ , \ d_{pitch} = \frac{2 \cdot T_{orque}}{F_T}$$

Radial Force

$$F_R = F_T \tan\left(\alpha_{pressure}\right)$$

Given Parameters

Gear Ratio

$$G_R = \frac{24}{9} = 2.666666667$$

Output Tourque (in-lbf)

$$T_O = 1.25$$

Input Shaft Diameter (Pinion, Power Transmission)

$$D_I = 0.125$$

Output Shaft Diameter (Gear, Short Shaft)

$$D_O = 0.25$$

Shaft Length (Both)

$$L = 1.0$$

Output Speed: 1100 rpm

$$N_O = 1\,100$$

Reference Parameters

Maximum Torsional Stress (psi) *(Machineries Handbook page 295)*

$$S_{Smax} = 8\,000$$

Maximum Bending Stress (psi) *(Machineries Handbook page 295)*

$$S_{max} = 16\,000$$

Pressure Angle (deg) (From McMasterCarr Detail for assigned Gears)

$$\alpha = 20$$

Input Pitch Diameter (in.)

$$d_{PI} = 0.375$$

Output Pitch Diameter (in.)

$$d_{PO} = 1.00$$

Calculations

Shaft Strength

Input Speed

$$N_I = G_R N_O = 2933.333333$$

Output Power (Hp)

$$P_O = \frac{D_O^3 N_O}{38} = 0.4523026316$$

Input Power (Hp)

$$P_I = \frac{D_I^3 N_I}{80} = 0.07161458333$$

Input Torque (in.-lbf)

$$T_I = \frac{T_O}{G_R} = 0.46875$$

Section Moduli (cubic in)

Input

$$Z_{PI} = \frac{\pi}{16} D_I^3 = 3.83495197 \times 10^{-4}$$

Output

$$Z_{PO} = \frac{\pi}{16} D_O^3 = 0.003067961576$$

Torsional Shearing Stresses (psi)

Input

$$S_{SI} = \frac{T_I}{Z_{PI}} = 1222.309963$$

Input Stress Safety Factor

$$S_{afetyI} = \frac{S_{Smax}}{S_{SI}} = 6.544984695$$

Output

$$S_{SO} = \frac{T_O}{Z_{PO}} = 407.4366543$$

Output Stress Safety Factor

$$S_{afetyO} = \frac{S_{Smax}}{S_{SO}} = 19.63495408$$

Both Shafts have sufficient diameters.

Bearing Selection and Design

INPUT

Input Shaft Tangential Force(lbf)

$$F_{TI} = \frac{2T_I}{d_{PI}} = 2.5$$

Input Shaft Radial Force (lbf)

$$F_{RI} = F_{TI} \tan(\alpha) = 5.592902361$$

OUTPUT

Output Shaft Tangential Force(lbf)

$$F_{TO} = \frac{2T_O}{d_{PO}} = 2.5$$

Output Shaft Radial Force (lbf)

$$F_{RO} = F_{TO}\tan(\alpha) = 5.592902361$$

Bearing Speeds

Input SFM

$$V_{SI} = \frac{D_I N_I}{3.82} = 95.98603839$$

Output SFM

$$V_{SO} = \frac{D_O N_O}{3.82} = 71.9895288$$

Input Bearing Length (in.)

$$L_{BI} = .25$$

Output Bearing Length (in.)

$$L_{BO} = 0.25$$

Bearing Material : PEEK

Maintenance Free is the way to be and you can get any color you want as long as it's black

Projected Area (sq. in.)

Input

$$A_I = D_I L_{BI} = 0.03125$$

Output

$$A_O = D_O L_{BO} = 0.0625$$

Pressure_Nominal (psi)

Input

$$p_I = \frac{F_{RI}}{A_I} = 178.9728755$$

Output

$$p_O = \frac{F_{RO}}{A_O} = 89.48643777$$

Pressure_Max

$$P_{max} = 4500$$

Velocity_Max

$$V_{max} = 400$$

PV Max

$$M_{AXPV} = 25000$$

PV Inupt

$$Q_I = p_I V_{SI} = 17178.8973$$

PV Output

$$Q_O = p_O V_{SO} = 6442.086489$$